UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450
EXAMINATION	:	NOVEMBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the ICT450 mock paper (November 2026). Answers are unofficial model answers, not official UTM marking schemes.

2. Marking points state the expected answer and the allocation of marks; every question retains its source paper and page reference.

3. Five questions x 20 marks; total = 100 marks.

4. Real-bank questions: 5. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 12 unique ICT450 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages.

© Hak Cipta Universiti Teknologi MARA

MARKING GUIDE

QUESTION 1 (20 marks)

Source: ICT450-2025-FEB · page [2] · bank id ICT450-2025-FEB-Q1

Reference answer

(a) Four data items for an online store. [4 marks]

- Customer name, address, phone and e-mail.
- Product ID, name, description and price.
- Order ID, order date, quantity and total.
- Payment method and delivery status.

(b) Two differences between a database and a spreadsheet. [4 marks]

- A database enforces relationships, keys and constraints; a spreadsheet stores independent cells and formulas.
- A database supports multi-user concurrency, security and recovery; a spreadsheet is mainly single-user and has limited integrity controls.

(c) Insert, update and delete rules of referential integrity. [6 marks]

- Insert rule: a child row cannot be inserted unless its foreign-key value matches an existing parent key or is NULL.
- Update rule: a parent key cannot be changed if dependent child rows would be orphaned, unless the change cascades.
- Delete rule: a parent row cannot be deleted while dependent child rows exist, unless the delete cascades or sets NULL.
- Example: an order cannot reference a customer that does not exist.

(d) Three differences between centralized and decentralized design. [6 marks]

- Data location: one site versus multiple sites.
- Control and administration: central versus local.
- Availability and cost: centralized is simpler but has a single point of failure; decentralized is more resilient but harder to coordinate.

Marking guide

- a) 1 mark for each of four data items (4).
- b) 2 marks for each difference (4).
- c) 2 marks for each referential-integrity rule with example (6).
- d) 2 marks for each difference (6).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 1 DATABASE CONCEPT (1).pdf; TOPIC 2 DATA MODEL (1).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

© Hak Cipta Universiti Teknologi MARA

QUESTION 2 (20 marks)

Source: ICT450-2018-DEC · page [2] · bank id ICT450-2018-DEC-Q2

Reference answer

(a) Business rules and three essentials. [8 marks]

- A business rule is a precise statement that describes a policy, constraint or relationship in the organisation's operations and must be reflected in the database.
- Essential 1: the rule must be precise and unambiguous, so every reader derives the same entities, attributes and constraints.
- Essential 2: it must be documented and traceable to its source (policy, procedure, form or interview).
- Essential 3: it must be enforceable in the design through keys, cardinalities, domains or check constraints.

(b) Importance of an index. [2 marks]

- An index provides a fast access path to rows, speeding up searches, joins and sorting; a unique index also enforces uniqueness.

(c) Four common sources of database failure. [4 marks]

- Hardware failure (disk, memory, power).
- Software failure (DBMS or application bug).
- Human error (accidental deletion or bad update).
- Natural disaster, network failure or malicious attack.

(d) Three backup levels. [6 marks]

- Full backup — a complete copy of the database.
- Differential backup — changes since the last full backup.
- Transaction-log backup — the log of individual transactions, allowing point-in-time recovery.

Marking guide

- a) 2 marks for the definition and 2 marks for each of three essentials (2 + 6 = 8).
- b) 2 marks for the indexing benefit.
- c) 1 mark for each of four sources (4).
- d) 2 marks for each of three backup levels (6).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 2 DATA MODEL (1).pdf; TOPIC 6 DATABASE DESIGN (1).pdf

© Hak Cipta Universiti Teknologi MARA

QUESTION 3 (20 marks)

Source: ICT450-2022-FEB · page [4, 5] · bank id ICT450-2022-FEB-Q3

Reference answer

Table 4 is a hotel reservation form with repeating menu lines and repeated customer and staff data.

(a) Normal form of Table 4. [3 marks]

- The table is in 1NF when flattened to one row per reservation menu line; each cell is atomic.
- It is not in 2NF because customer, staff and menu details depend on parts of the composite key.
- It is not in 3NF because MenuName and TypeOfCuisine depend on MenuID, which is not the whole key.

(b) Relational schema, keys and dependencies. [6 marks]

Flattened relation.

```
RESERVATION (CustomerID, ReservationID, ReservationDate, TimeSlot, FirstName, LastName, Email,
PhoneNumber, NumberOfGuests, StaffID, StaffName, CuisineType, MenuID, MenuName)
```

Functional dependencies.

```
CustomerID → FirstName, LastName, Email, PhoneNumber
StaffID → StaffName
MenuID → MenuName, CuisineType
ReservationID → ReservationDate, TimeSlot, CustomerID, NumberOfGuests, StaffID
ReservationID, MenuID → (reservation menu line)
```

(c) Normalise to 3NF. [11 marks]

1NF: one row per reservation-menu line; key (ReservationID, MenuID).

2NF: remove customer, staff and menu partial dependencies.

2NF/3NF relations.

```
CUSTOMER (CustomerID, FirstName, LastName, Email, PhoneNumber)
STAFF (StaffID, StaffName)
RESERVATION (ReservationID, ReservationDate, TimeSlot, CustomerID, NumberOfGuests, StaffID)
MENU (MenuID, MenuName, CuisineType)
RESERVATION_MENU (ReservationID, MenuID)
```

3NF: MenuName and CuisineType depend on MenuID, so MENU is already separated; no further transitive dependency remains.

Sample 3NF data: CUSTOMER

CustomerID	FirstName	LastName	Email	Phone
A01	Intan	Nasa	intan@gmail.com	017-12345678
A02	Fatin	Ruslan	fatin@gmail.com	

Sample 3NF data: RESERVATION

ReservationID	Date	TimeSlot	CustomerID	Guests	StaffID
R1	19/02/2022	8:00 PM	A01	2	111
R3	20/02/2022	8:00 PM	A01	3	222

Marking guide

- a) 3 marks: 1NF and reasons why not 2NF/3NF.
- b) 6 marks: schema (2), key (2), dependencies (2).
- c) 11 marks: 1NF (2), 2NF (4), 3NF (5).

Provenance: model_unofficial · confidence medium · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 5 NORMALIZATION (1).pdf

Assumptions / notes

- The menu field is multi-valued; the reservation key is (ReservationID, MenuID). Staff and customer names depend on their IDs.

© Hak Cipta Universiti Teknologi MARA

QUESTION 4 (20 marks)

Source: ICT450-2024-JAN · page [5] · bank id ICT450-2024-JAN-Q4

Reference answer

Movie database schema.

```
MOVIE (MovieID, MovieName, Genre, DateRelease, NetProfit, DirectorID*)
DIRECTOR (DirectorID, DirectorName, Nationality)
REVIEW (ReviewID, Description, MovieID*, ReviewerID*)
REVIEWER (ReviewerID, ReviewerName)
```

(a) Movies released from 2021 until 2023. [4 marks]

```
SELECT *
FROM MOVIE
WHERE DateRelease BETWEEN DATE '2021-01-01' AND DATE '2023-12-31';
```

(b) Total movies directed by Malaysian directors. [4 marks]

```
SELECT COUNT(*) AS total_movies
FROM MOVIE m
JOIN DIRECTOR d ON d.DirectorID = m.DirectorID
WHERE d.Nationality = 'Malaysian';
```

(c) Average net profit for each genre. [4 marks]

```
SELECT Genre, AVG(NetProfit) AS "Average Net Profit"
FROM MOVIE
GROUP BY Genre;
```

(d) Reviewers with more than two reviews. [4 marks]

```
SELECT r.ReviewerName
FROM REVIEWER r
JOIN REVIEW v ON v.ReviewerID = r.ReviewerID
GROUP BY r.ReviewerID, r.ReviewerName
HAVING COUNT(*) > 2;
```

(e) Movie that received the most reviews. [4 marks]

```
SELECT m.MovieName
FROM MOVIE m
JOIN REVIEW v ON v.MovieID = m.MovieID
GROUP BY m.MovieID, m.MovieName
ORDER BY COUNT(*) DESC
FETCH FIRST 1 ROW ONLY;
```

MySQL: replace FETCH FIRST 1 ROW ONLY with LIMIT 1.

Marking guide

- a) 4 marks: date range and SELECT.
- b) 4 marks: movie-director join and Malaysian filter.
- c) 4 marks: AVG, GROUP BY genre and alias.
- d) 4 marks: join, GROUP BY reviewer, HAVING > 2.
- e) 4 marks: join, COUNT, descending order and top-1.

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.

Knowledge base basis: TOPIC 7 SQL.pdf

Assumptions / notes

- DateRelease is stored as a date; 'most reviews' is answered with a top-1 ordered query.

© Hak Cipta Universiti Teknologi MARA

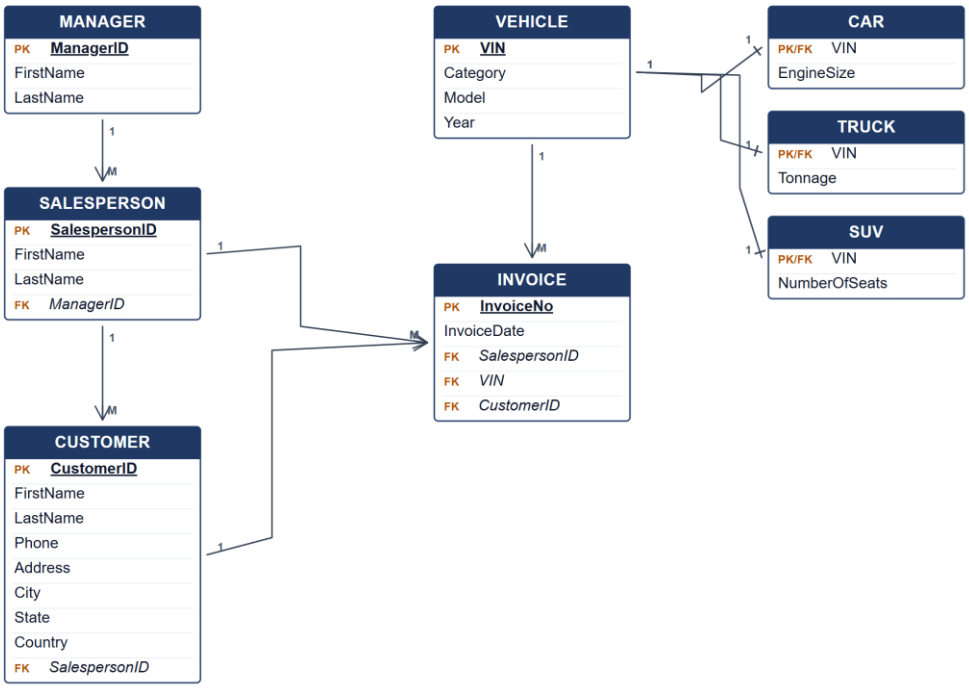
QUESTION 5 (20 marks)

Source: ICT450-2019-DEC · page [5] · bank id ICT450-2019-DEC-Q5

Reference answer

The ERD models managers, salespersons, customers, vehicles and invoices. Vehicle is specialised into Car, Truck and SUV.

(a) Construct the ERD with Crow's Foot notation. [16 marks]



Car Dealership System ERD.

(b) Two reports from the ERD. [4 marks]

- Salesperson performance — invoices and sales value per salesperson, grouped by manager.
- Vehicle category report — number and value of cars, trucks and SUVs sold.

Marking guide

- a) 16 marks: manager/salesperson/customer (5); vehicle and three subtypes (5); invoice and relationships (4); keys (2).
- b) 2 marks for each report (4).

Provenance: model_unofficial · confidence medium · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 4 ERD_stu (2).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- A vehicle is one of Car, Truck or SUV, and an invoice records one vehicle sale.

© Hak Cipta Universiti Teknologi MARA